If the above two conditions are not satisfied, then $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents,

(i) a parabola if $h^2 - ab = 0$

(ii) an ellipse if $h^2 - ab < 0$

(iii) a hyperbola if $h^2 - ab > 0$

In this chapter, we study about parabola only.

3.5.1 Parabola

Definition 3.4

The locus of a point whose distance from a fixed point is equal to its distance from a fixed line is called a parabola.

$y^2 = 4ax$ is the standard equation of the parabola. It is open rightward.

3.5.2 Definitions regarding a parabola: $y^2 = 4ax$

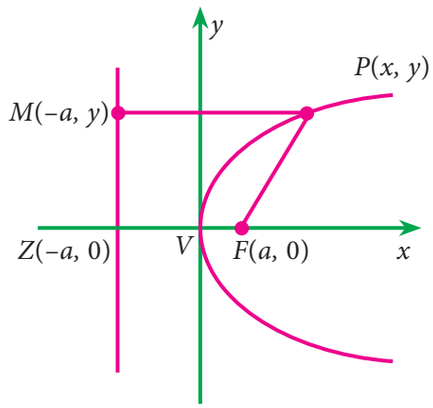


Fig. 3.10

Focus	The fixed point used to draw the parabola is called the focus (F). Here, the focus is $F(a, 0)$.
Directrix	The fixed line used to draw a parabola is called the directrix of the parabola. Here, the equation of the directrix is $x = -a$.
Axis	The axis of the parabola is the axis of symmetry. The curve $y^2 = 4ax$ is symmetrical about x-axis and hence x-axis or $y = 0$ is the axis of the parabola $y^2 = 4ax$. Note that the axis of the parabola passes through the focus and perpendicular to the directrix.
Vertex	The point of intersection of the parabola with its axis is called its vertex. Here, the vertex is $V(0, 0)$.
Focal distance	The distance between a point on the parabola and its focus is called a focal distance
Focal chord	A chord which passes through the focus of the parabola is called the focal chord of the parabola.
Latus rectum	It is a focal chord perpendicular to the axis of the parabola. Here, the equation of the latus rectum is $x = a$. Length of the latus rectrum is $4a$.

3.5.3 Other standard parabolas

1. Open leftward : $y^2 = -4ax[a > 0]$

If $x > 0$, then y become imaginary. i.e., the curve exist for $x \leq 0$.

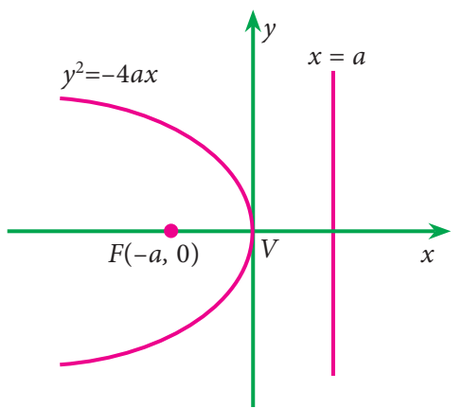


Fig. 3.11

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